

INSTITUTIONAL EQUITY EXECUTION & TRADING ANALYTICS

# Transaction Cost Analysis (TCA) Executive Performance Report

Execution Quality, Slippage Attribution, Broker League Tables & Smart Routing Optimization

## DATASET & PORTFOLIO METADATA

- Execution Sample: 100,000 Institutional Equity Orders | Period: FY2025 (Jan 2025 - Dec 2025)
- Total Notional Traded: \$95.86 Billion USD | Total Execution Friction: \$13.01 Million USD (0.99 bps)
- Benchmarks Evaluated: Arrival Price Slippage (0.58 bps) & Interval VWAP Slippage (0.45 bps)
- Stakeholder Audience: Chief Investment Officer, Head of Trading, Execution Committee

# Executive Scorecard: FY2025 Execution Performance at a Glance

Total Traded Notional

\$95.86B

100,000 orders executed

Total Execution Drag

\$13.01M

0.99 bps effective cost

Avg Arrival Slippage

0.58 bps

Market impact vs. decision

Avg VWAP Slippage

0.45 bps

Volume tracking efficiency

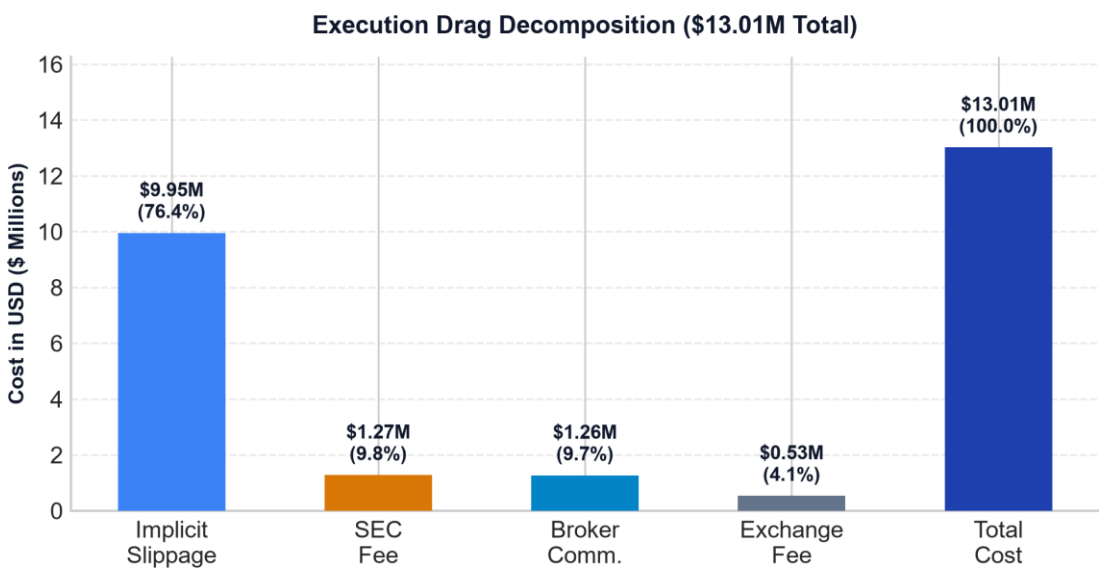
Order Fill Rate

99.25%

97.12% full, 2.88% partial

COST DECOMPOSITION

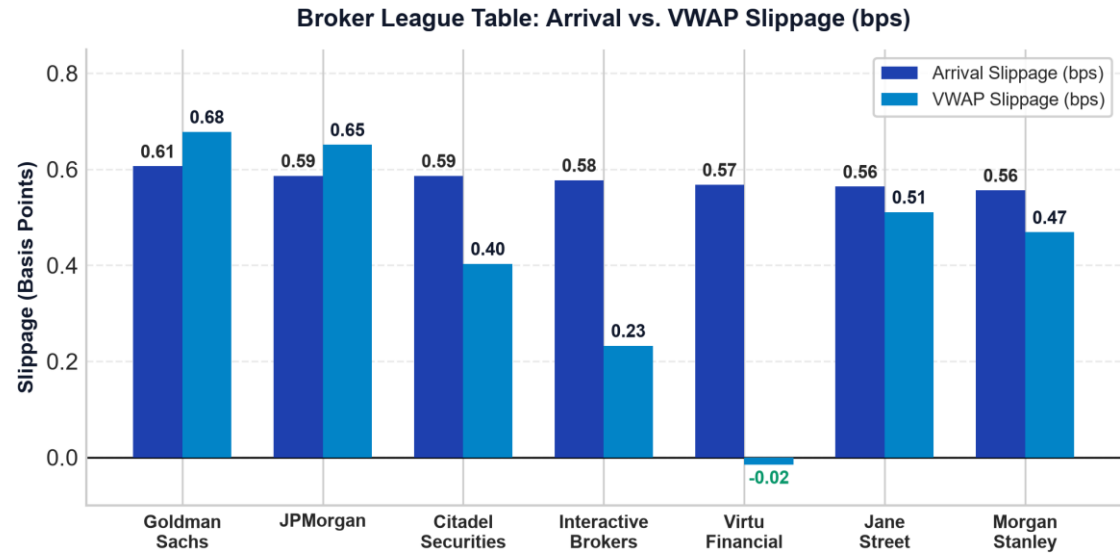
## Cost Anatomy: Implicit Market Drag & Buy/Sell Asymmetry



IMPLICIT VS. EXPLICIT COST DYNAMICS

- Implicit Cost (\$9.95M / 76.4%): Arrival price slippage (0.58 bps avg) represents temporary and permanent price distortion caused by order execution pace.
- Commission (\$1.26M / 9.7%): Averages 0.13 bps across desks. Highly competitive, but broker tiering should be tied to execution quality rather than flat rate.
- SEC Fees (\$1.27M / 9.8%): Statutory fee passed directly through from executing brokers.
- Exchange / Take Fees (\$0.53M / 4.1%): Incurred primarily by lit venue aggressive routing. Routinely avoidable through smart dark pegging.

## Executing Broker Benchmarking: League Table & Slippage Efficiency

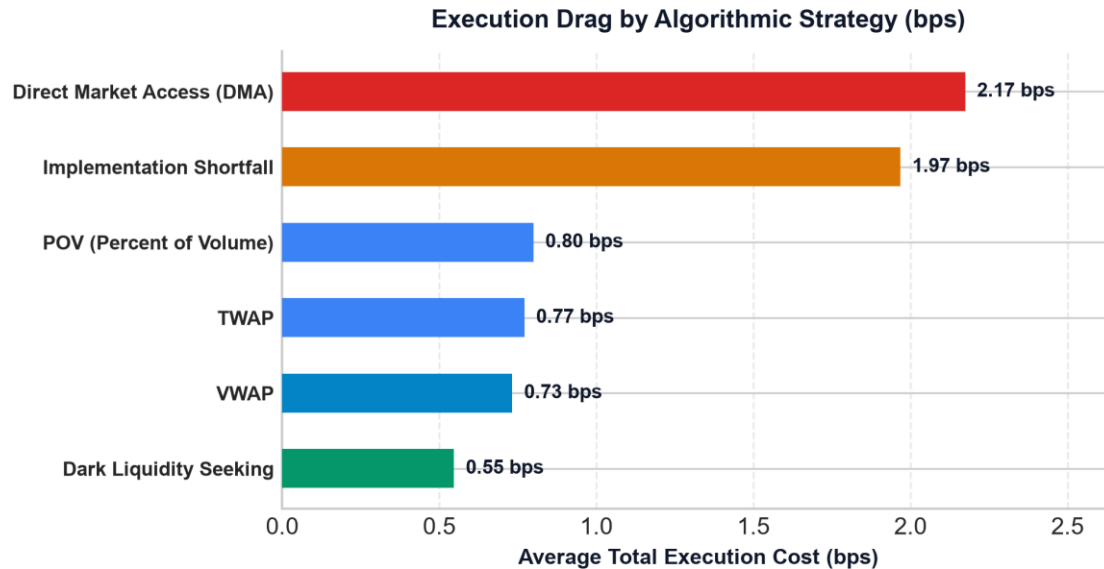


| Broker              | Notional | Arr. (bps) | VWAP (bps) | Cost (bps) | Cost/Ord |
|---------------------|----------|------------|------------|------------|----------|
| Morgan Stanley      | \$21.27B | 0.56       | 0.47       | 0.97       | \$123.52 |
| Goldman Sachs       | \$18.85B | 0.61       | 0.68       | 1.02       | \$133.78 |
| JPMorgan            | \$17.11B | 0.59       | 0.65       | 1.00       | \$130.36 |
| Virtu Financial     | \$14.65B | 0.57       | -0.02      | 0.98       | \$129.84 |
| Citadel Securities  | \$11.43B | 0.59       | 0.40       | 1.00       | \$134.84 |
| Jane Street         | \$7.66B  | 0.56       | 0.51       | 0.98       | \$132.93 |
| Interactive Brokers | \$4.89B  | 0.58       | 0.23       | 0.99       | \$129.46 |

### MANAGEMENT TAKEAWAYS ON BROKER PERFORMANCE

- **Top Flow & Scale Efficiency:** Morgan Stanley captured the largest share (\$21.27B / 22.0k orders) while maintaining the lowest cost per order (\$123.52) and total cost (0.97 bps).
- **Benchmark Outperformer:** Virtu Financial delivered an outstanding -0.02 bps VWAP slippage, proving superior execution against volume schedules.
- **Underperformance Flag:** Goldman Sachs and JPMorgan exhibit elevated VWAP slippage (0.68 bps and 0.65 bps) and highest total cost (1.02 bps and 1.00 bps). Recommend placing under-allocated flow on conditional review.

## Algorithmic Strategy Performance: Cost, Market Impact & Suitability



### ALGORITHMIC PROFILES & RECOMMENDATIONS

#### Dark Liquidity Seeking (0.55 bps | \$551.8k Cost)

- Best performing strategy desk-wide. Arrival slippage is only 0.13 bps.
- Minimizes market footprint by crossing mid-point in dark venues without displaying quotes.
- Action: Expand mandate for all liquid names and patient institutional orders.

#### VWAP & TWAP (0.73 - 0.77 bps | \$5.11M Combined Cost)

- Represents 48% of total desk notional (\$46.2B). Near-zero VWAP slippage (-0.01 bps).
- Excellent benchmark compliance across large-cap holdings (MSFT, AAPL, NVDA).
- Action: Maintain as default core schedule algorithm.

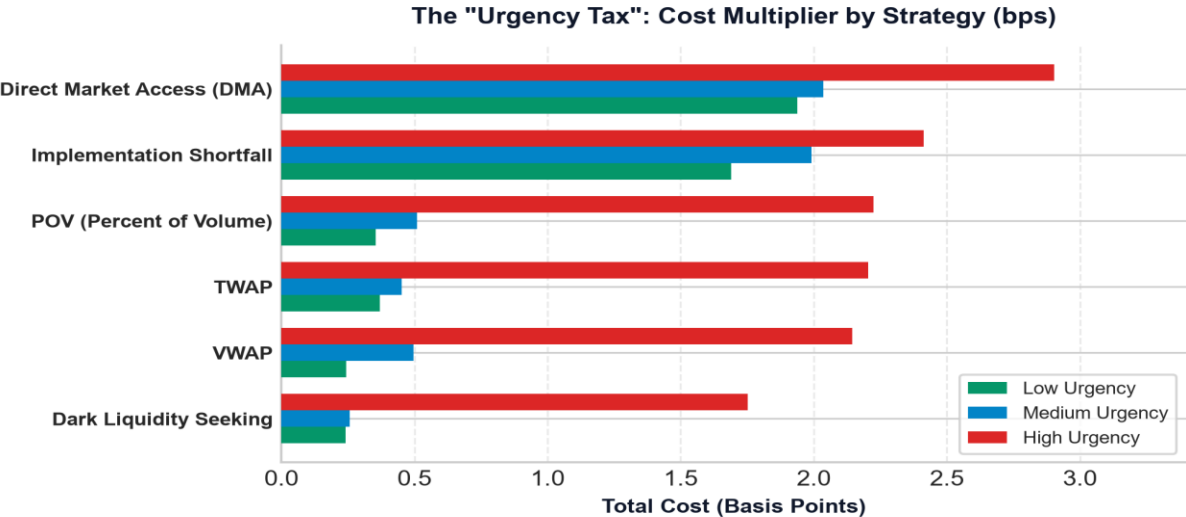
#### POV (Percent of Volume) (0.80 bps | \$2.29M Cost)

- 0.38 bps arrival slippage. Dynamically adapts to intraday volume surges.
- Action: Ideal for medium-cap stocks with unpredictable intraday participation.

#### DMA & Implementation Shortfall (1.97 - 2.17 bps | \$5.07M Cost)

- Severe cost penalties: DMA averages 2.17 bps (1.75 bps arrival, 2.04 bps VWAP).
- High market footprint and immediate adverse selection.
- Action: Strictly restrict DMA to specialist high-urgency desk overrides with mandatory justification.

# The 'Urgency Tax': Quantifying Market Impact & Order Pace

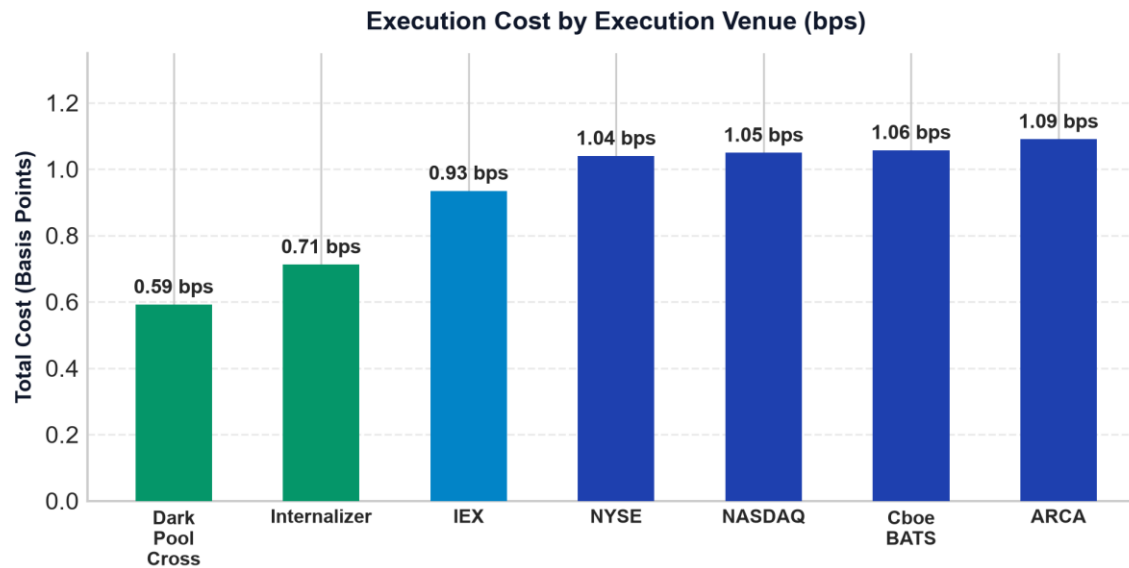


| Algo Strategy              | Low (bps) | Medium (bps) | High (bps) |
|----------------------------|-----------|--------------|------------|
| Dark Liquidity Seeking     | 0.24      | 0.26         | 1.75       |
| VWAP                       | 0.24      | 0.50         | 2.14       |
| POV (Percent of Volume)    | 0.35      | 0.51         | 2.22       |
| TWAP                       | 0.37      | 0.45         | 2.20       |
| Implementation Shortfall   | 1.69      | 1.99         | 2.41       |
| Direct Market Access (DMA) | 1.94      | 2.04         | 2.90       |

## ORDER URGENCY SCORECARD

- High Urgency: 19,835 orders (19.8% flow) consume \$5.53M (42.5% of total desk cost) at 2.21 bps average.
- Medium Urgency: 45,180 orders (45.2% flow) consume \$4.70M (36.1% of cost) at 0.77 bps average.
- Low Urgency: 34,985 orders (35.0% flow) consume \$2.79M (21.4% of cost) at 0.59 bps average.
- The Multiplier: Shifting from Low to High urgency expands execution cost by 3.74x

## Liquidity Venues: Lit Exchanges vs. Alternative Dark Pools



### VENUE MICROSTRUCTURE & ROUTING ARBITRAGE

#### Lit Exchanges (NASDAQ, NYSE, Cboe BATS, ARCA)

- Volume Share: 80.3% (\$76.98B notional) | Total Cost: \$11.05M
- Average Cost: 1.04 to 1.09 bps. ARCA is the most expensive lit venue at 1.09 bps.
- Microstructure Issue: Subject to exchange take fees, quote fading, and high-frequency front-running.

#### IEX (Investors Exchange) (0.93 bps | \$7.44B Notional)

- 0.12 - 0.16 bps cheaper than NASDAQ and NYSE.
- The 350-microsecond 'speed bump' prevents latency arbitrage and crumb routing, delivering measurably cleaner fills.

#### Internalizes (0.71 bps | \$4.88B Notional)

- High execution quality, particularly via Interactive Brokers (\$0.08 bps internalizer cost).
- Provides substantial price improvement against NBBO without market impact.

#### Dark Pool Cross (0.59 bps | \$6.56B Notional)

- 44% cost reduction relative to lit exchanges (0.59 bps vs. 1.05 bps).
- Current desk utilization is only 6.8% of total volume.
- Primary Opportunity: Shifting 10-15% more volume to dark pools represents ~\$600k-\$900k in frictionless savings.

## Strategic Action Plan: Execution Optimization & \$1.75M Savings Roadmap

### 1. Enforce Urgency Governance & Desk Tiering

- Finding: High urgency orders drive 42.5% of costs (\$5.53M) at 2.21 bps.
- Action: Implement pre-trade approval threshold for 'High Urgency' flags over \$5M notional. Require PM sign-off.
- Target: Convert 30% of High Urgency flow to Medium/Low scheduled algorithms.
- Estimated Annual Savings: \$650,000 - \$850,000

### 2. Optimize Smart Order Routing (SOR) & Dark Liquidity

- Finding: Dark Pools (0.59 bps) and Internalizes (0.71 bps) beat lit exchanges (1.05 bps) by >40%.
- Action: Reconfigure broker router parameters to mandate midpoint dark crosses prior to routing to lit books.
- Target: Expand off-exchange flow from 11.9% to 25% of total executed notional.
- Estimated Annual Savings: \$500,000 - \$750,000

### 3. Rationalize DMA Usage & Sunset Aggressive Defaults

- Finding: DMA generates 2.17 bps cost and 1.75 bps arrival slippage with minimal execution benefits.
- Action: Transition 50% of DMA volume into Dark Liquidity Seeking and customized VWAP/TWAP.
- Target: Cap desk-wide DMA volume at < 2.5% of total orders.
- Estimated Annual Savings: \$300,000 - \$450,000

### 4. Broker Scorecard Allocation & Commission Renegotiation

- Finding: Virtu delivered -0.02 bps VWAP slippage; Goldman Sachs and JPMorgan lagged peer averages (1.00-1.02 bps).
- Action: Allocate +15% order flow to top performers (Virtu, MS, IBKR). Use TCA metrics to renegotiate pricing with underperforming tiers.
- Target: Reduce blended commission from 0.13 bps to 0.10 bps.
- Estimated Annual Savings: \$250,000 - \$350,000